

Adjectival agreement is interpretable: Evidence from summative agreement in Mexican Spanish and Brazilian Portuguese*



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Bosque’s summative agreement puzzle

Premise: Adjectival agreement is not interpreted; it agrees with the syntactic number of a noun
Puzzle: Coordination of 2 singular postnominal adjectives can modify a plural noun (Bosque, 2006)

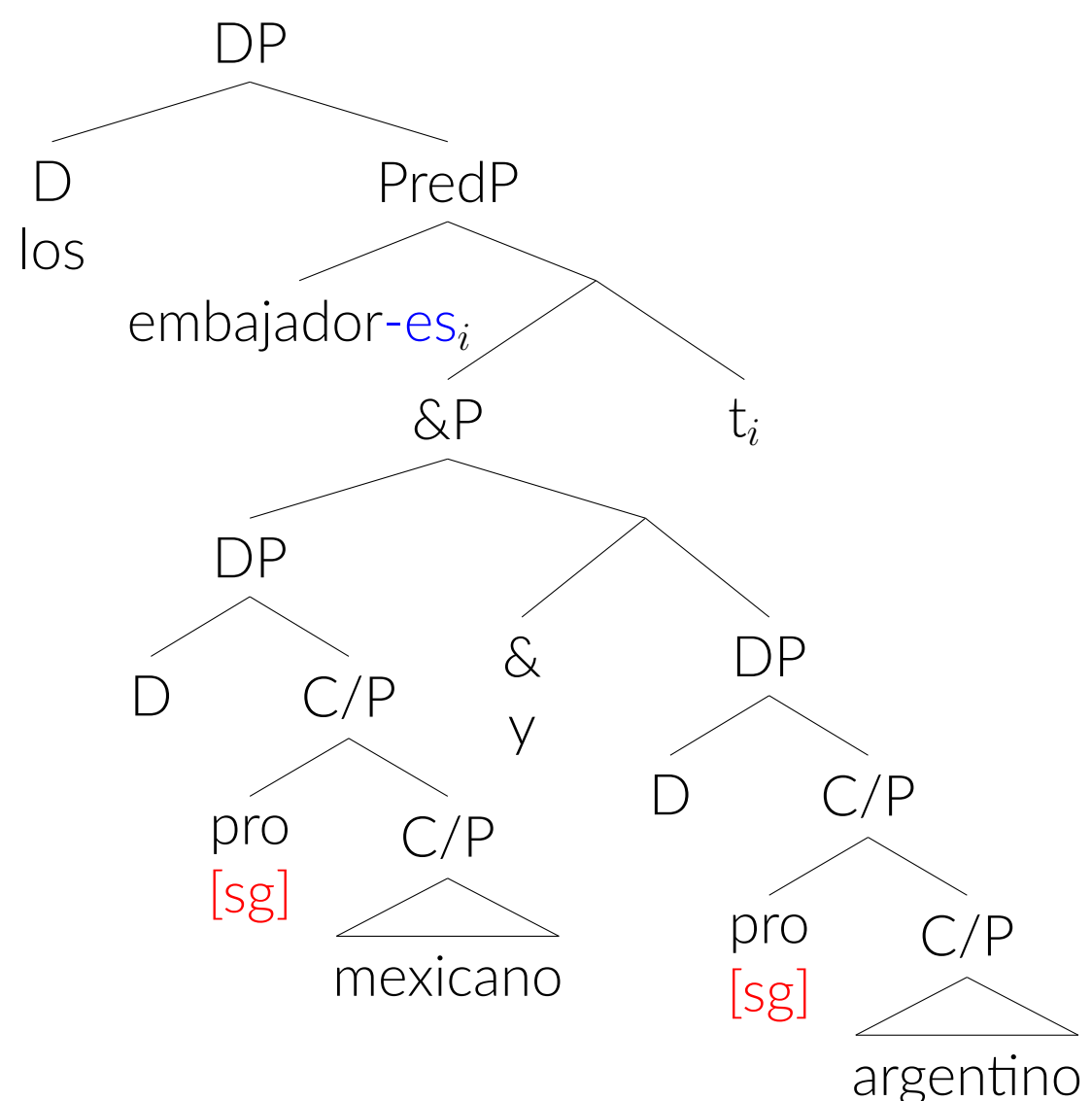
(1) Los embajadores mexicano y argentino vinieron a la conferencia
The ambassadors_{pl} Mexican_{sg} and Argentinian_{sg} came to the conference

Refers to two ambassadors (one Mexican, one Argentinian)

Past analyses: Adjectival agreement is not interpreted

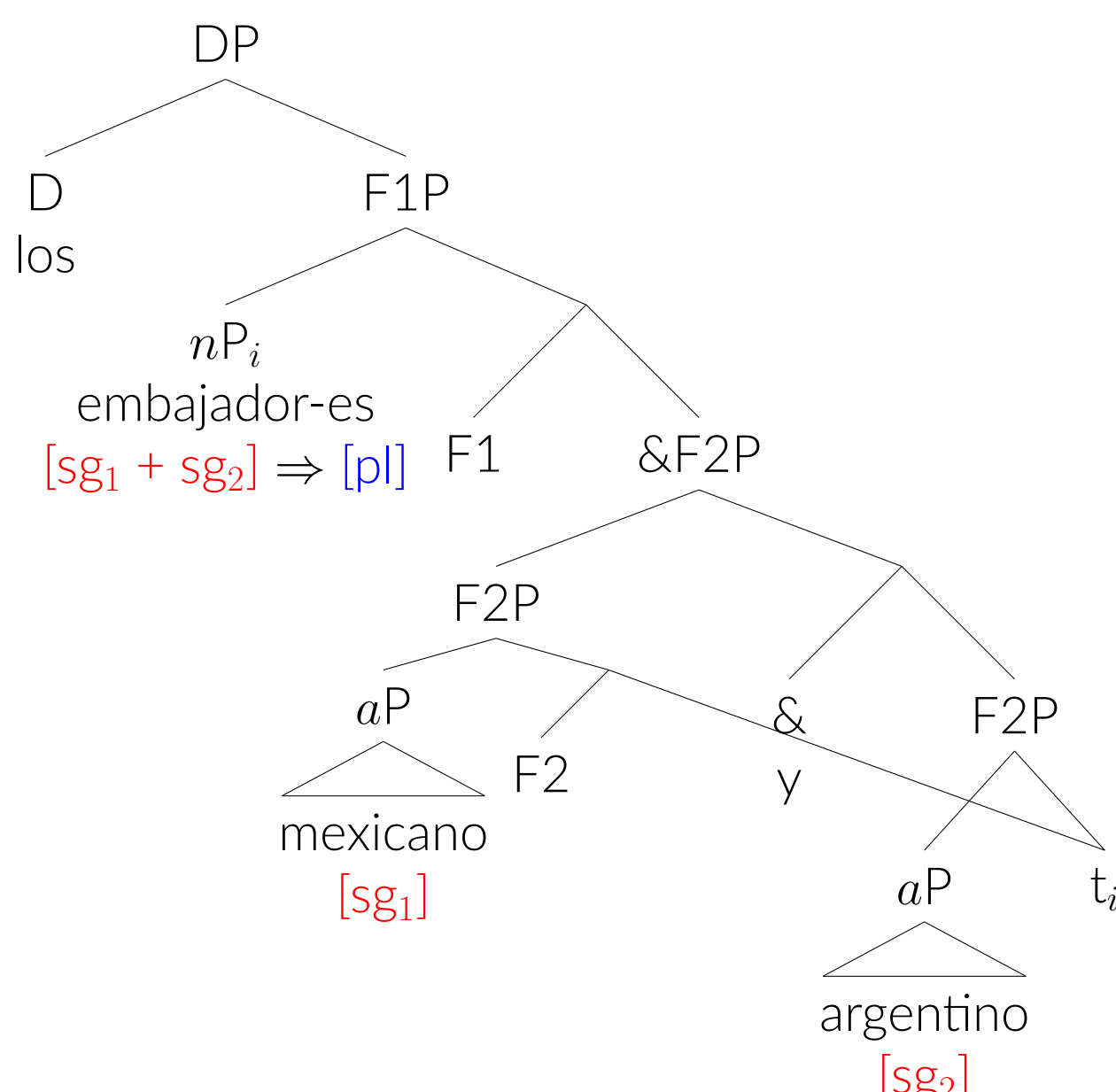
1. Bosque (2006): Small clause

- Two coordinated DPs with a [sg] *pro* subject triggering agreement in each adjective.
- Number is interpreted on each *pro* (not on the adjectives), the coordination requires [pl] on the matrix subject *embajadores*



2. Adamson (2025): Multidominance

- Each adjective agrees with a separate unpronounced [sg] feature on the noun.
- Morphology realizes the sum of the two [sg] features as a plural on the subject.



Both analyses:

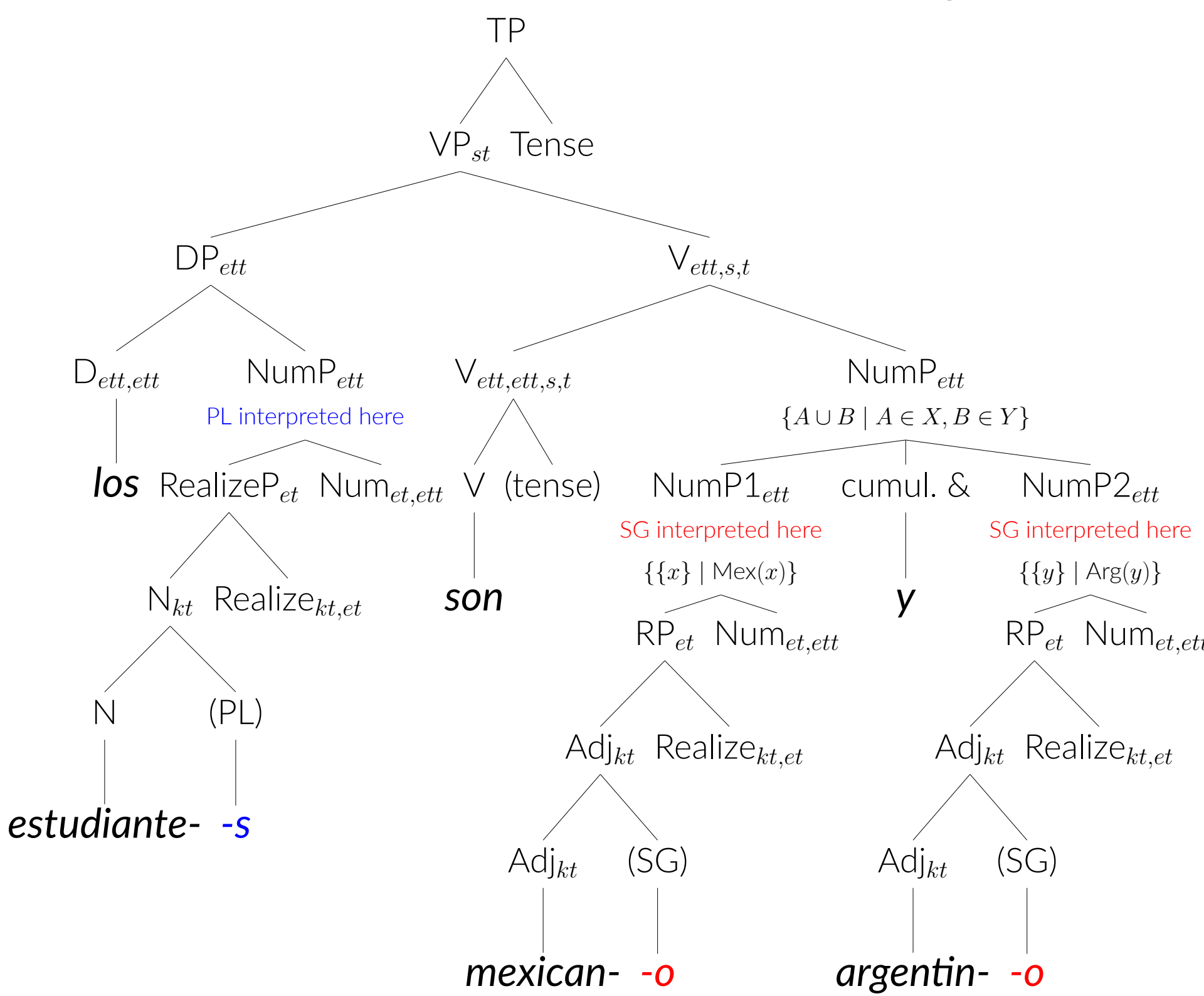
- Number is interpreted on *pro*/N, adjective number is not interpreted and agrees with it
- Summative agreement is restricted to ‘relational’ non-predicative denominal adjectives (cf. Bosque, 1993; Demonte, 1999), and not possible if there is no local noun

Prediction: Summative agreement not possible in predicative uses of coordinated adjectives

Our Hypothesis: Number on adjectives is interpreted

- SG/PL filter sets of sets:** SG only keeps singletons; PL only keeps those of cardinality > 1 (or keeps all if Sauerland et al. 2005 is correct about PL not being interpreted)
- SG/PL** cannot be interpreted where they attach; they are **interpreted higher**, where Number-head creates the semantic object they can apply to
- Compositional stashing:** If [z [xy]] can’t be interpreted, try [[zx]y] (allows PL/SG to ‘float up’ the tree until interpretable (similar effect to LF-movement within words proposed in Pesetsky 1985)¹
- Summative agreement reflects **cumulative predication**, derived by **cumulative and** (Krifka, 1990)

Copula (intersective, predicative use of summative agreement)



- Cumulative & derives set of binary sets, each with one Mexican and one Argentinian
- Copula: Asserts that subject is member of this set of sets
- (In order to analyze relational modification (see below), we assume nouns/adjectives underlyingly denote kinds/kind predicates, and are turned into *et* predicates by Realize operator (following McNally and Boleda (2004); Morzycki (2016), and sets of sets by Num (Num = *))

Prediction: Summative agreement also possible with intersective and predicative uses

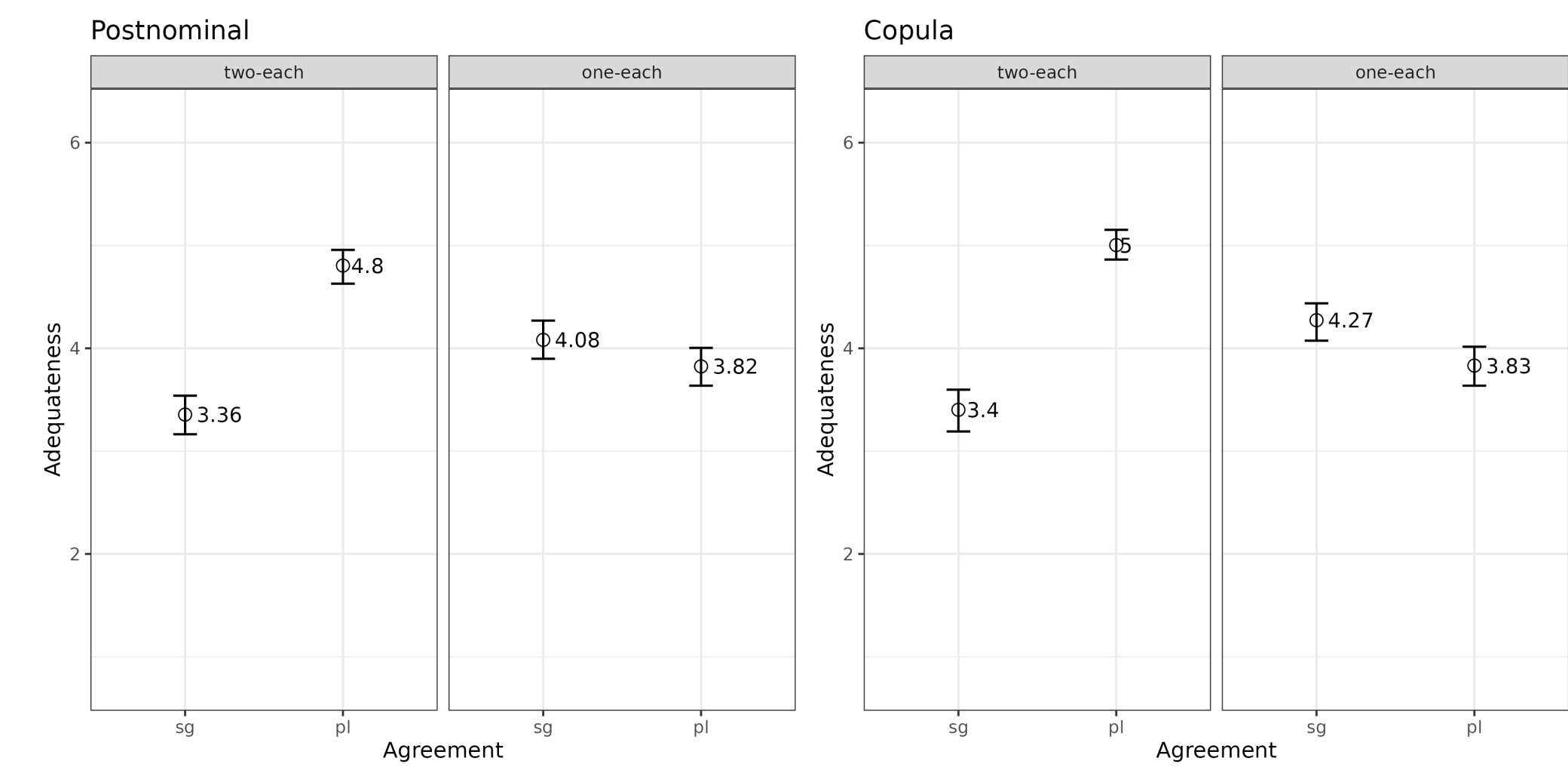
Rating experiment: Postnominal and predicative uses of summative agreement

32 participants from each language rated sentences (in their respective language) with one of two contexts (8 item sets for each copula and postnominal cases):

Context: We hired some students **[Two-each:]** Four of them start tomorrow. Two are Mexican, and two Argentinian; **[One-each:]** Two of them start tomorrow. One is Mexican, and the other Argentinian.

Question: Can you convey this information using this sentence? **[Postnominal:]** Los estudiantes mexicano(s) y argentino(s) vienen mañana; **[Copula:]** Los estudiantes son mexicano(s) y argentino(s).

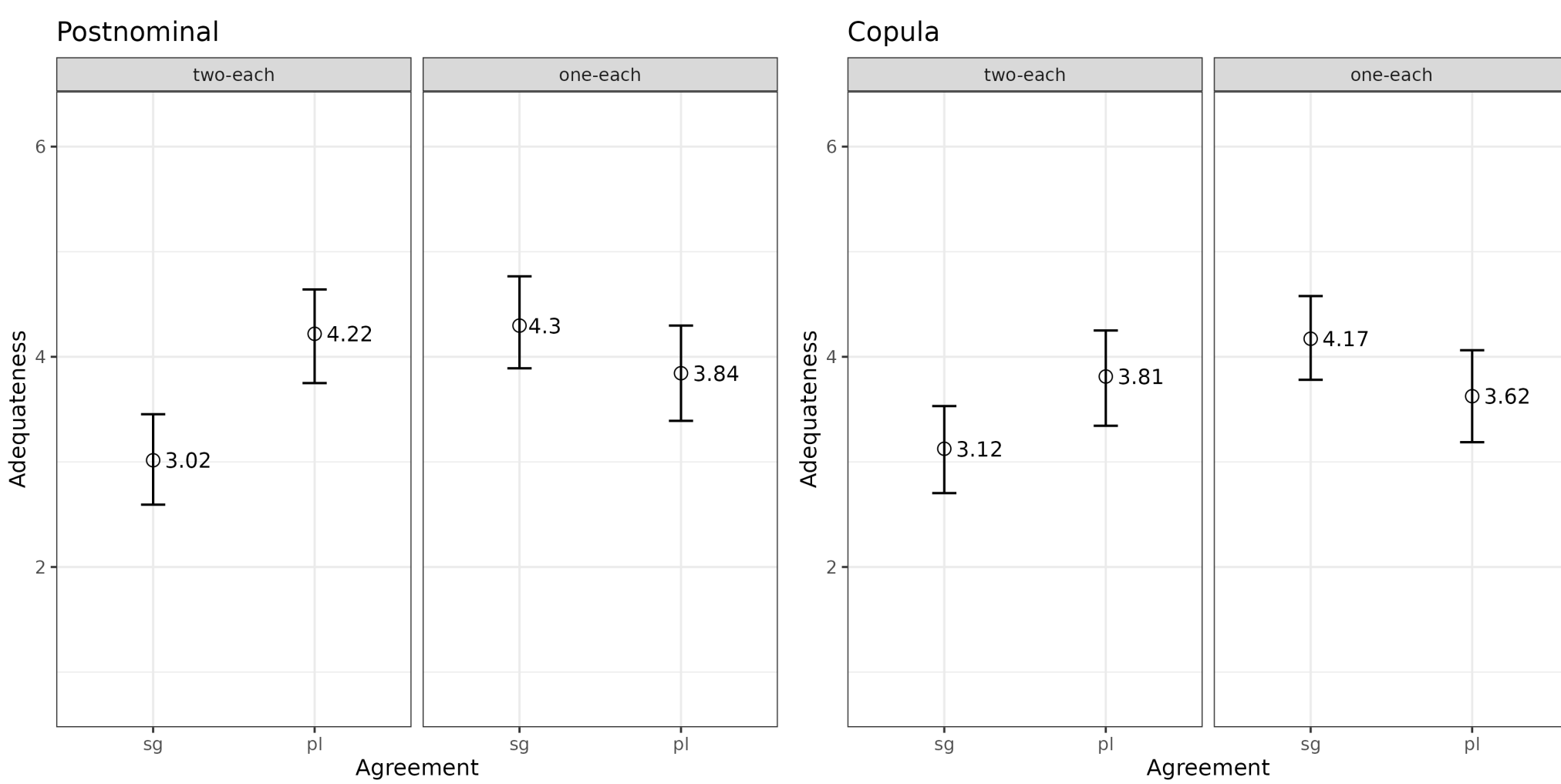
Mexican Spanish



Findings

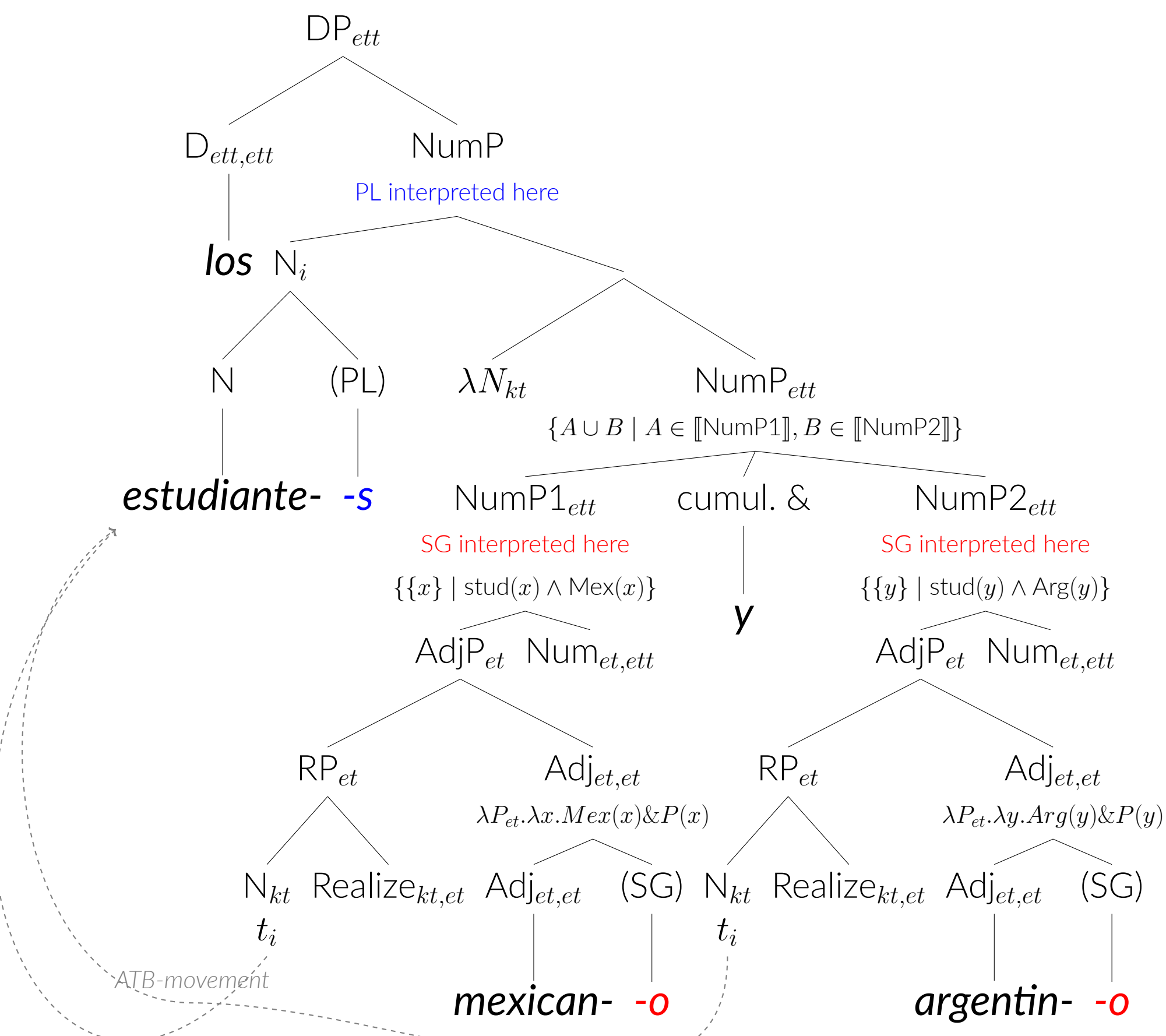
- Interaction: [sg] better in one-each condition, [pl] worse
- Summative agreement possible in copula case** (not expected by Bosque (2006) and Adamson (2025))
- [sg] lower rating in one-each compared to [pl] in two-each (frequency effect? [sg] agreement with [pl] noun is rare)

Brazilian Portuguese



Postnominal adjectives: Relational vs. intersective modification

Intersective modification



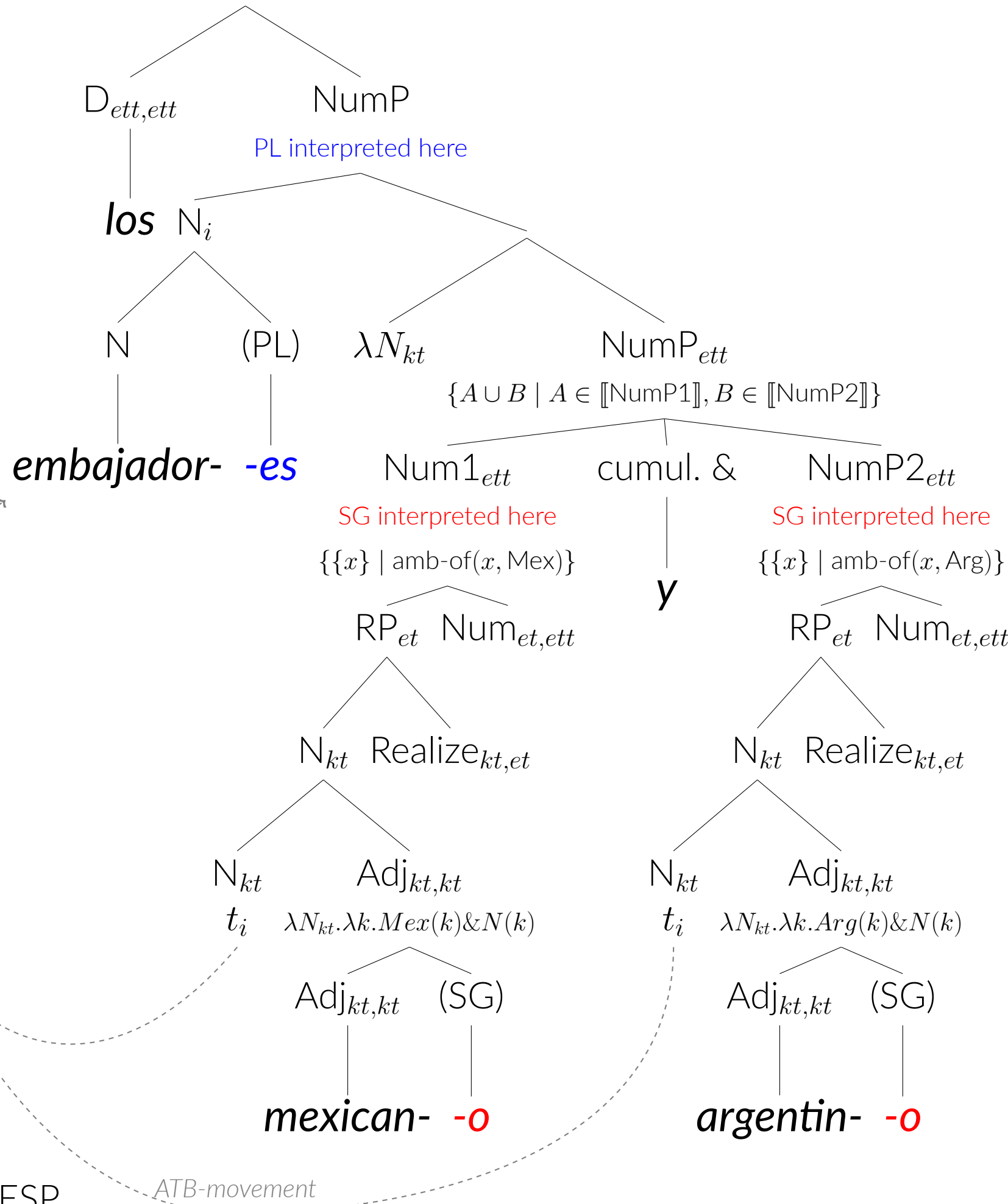
Intersective vs. Relational: Similarities

- Postnominal order derived by movement** in both cases, following Cinque et al. (2014)
- Noun shared by conjuncts** (similar to Adamson’s analysis, but instead of multidominance we assume ATB movement (cf. Harizanov and Gribanova, 2015))
- Modification is intersective in both cases** (intersective modification of kind in the case of relational modification, following McNally and Boleda 2004
- (*et*) meanings derived by **Realize** operator (Morzycki, 2016)
- Modification is below determiner and therefore **restrictive** in both cases (unlike the copula case, where restrictive modification not possible)

Intersective vs. Relational: Differences

- Syntactic/semantic:** In the intersective case, the adjectives scope above realize operator, resulting in predicate modification; in the relational case, they scope below, resulting in kind modification
- Pragmatic:** Accommodating that there is one each in the set (a precondition of summative agreement) is often trivial with relational modification (e.g. there is only one ambassador from each country), while the intersective case relies more on context (and can hence sound infelicitous out of context)

Relational modification



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¹An alternative could be that SG/PL lowers to the noun (Embick and Noyer, 2001)—the crucial assumption is just that SG/PL is not interpreted where they attach

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